Technical Proposal

Process Monitoring & Efficiency Improvement (PMEI) System

for

Cold Rolling Mill (Rewinding Line & 4-HI SPM),
Jindal Steel Odisha Limited (JSOL), Angul

(Ref No - 25/JSOL/PMEI/01/v0)
13th Nov. 2025 Ver. 2

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ABBREVIATIONS

PMEI: Process Monitoring & Efficiency Improvement system
JSOL: Jindal Steel Odisha Ltd.
HIL: Hitachi India Pvt. Ltd.
SPM: Skin Pass Mill
SV: Supervisor
HMI: Human Machine Interface
PLC: Programmable Logic Controller
L1: Level-1 system
L2: Level-2 system
L3: Level-3 system

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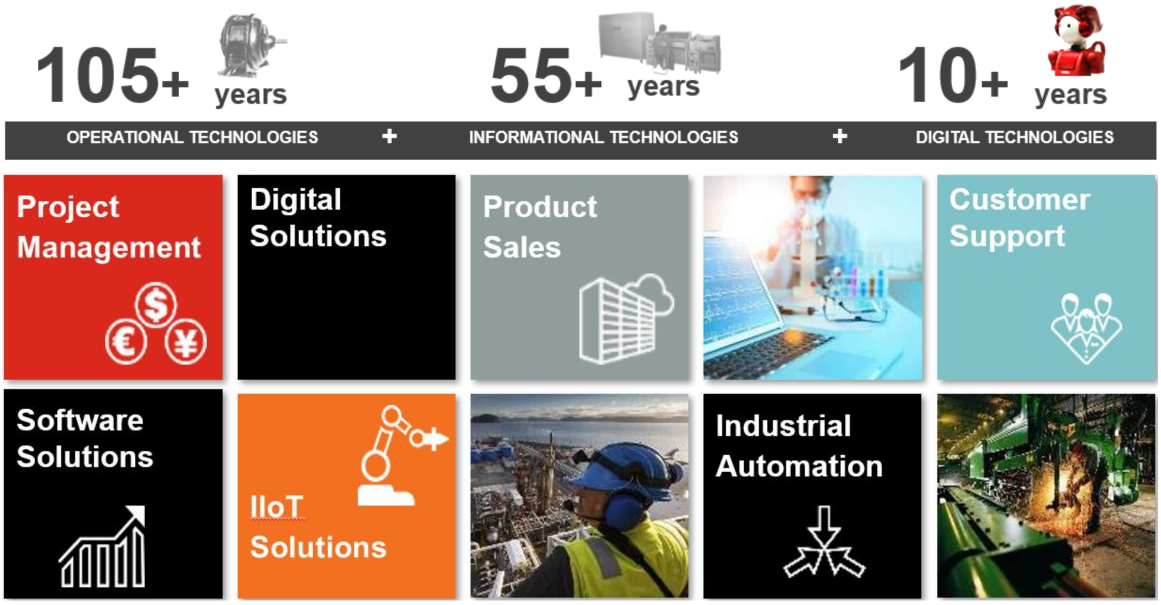
1. EXECUTIVE SUMMARY

Hitachi India Pvt. Ltd. (hereafter also referred to as HIL) provides automation solutions and digital solutions designed and developed for the industrial sector, particularly the metals industry. It takes advantage of more than 105 years of OT (Operation Technology) experience with process knowledge experts & SMEs and 55+ years of IT experience with the newest automation technology.



HIL is uniquely qualified to provide the control system and digital solutions you desire due to our in-house team of well-trained and certified professionals with the right experience & knowledge in designing, developing, and deploying the best solutions possible over a wide range of sectors & areas.

Some of our clients include:

TATA	JSW Steel	सेल SAIL
ADITYA BIRLA HINDALCO	AM/NS INDIA	posco

2. GENERAL OVERVIEW

This document is a technical proposal for design, engineering, supply & commissioning of Process Monitoring & Efficiency Improvement (PMEI) system at Cold Rolling Mill facility particularly for Rewinding Line & 4Hi Skin Pass Mill Line of Jindal Steel Odisha Limited, Angul, Odisha, India. This proposal is in response to the request received from the JSOL, dated Aug 7, 2025.

Hitachi India Pvt. Ltd. (hereafter SELLER) is pleased to submit this proposal for implementing PMEI system at Cold Rolling Mill facility particularly for Rewinding Line & 4Hi Skin Pass Mill Line of Jindal Steel Odisha Limited, India (hereafter BUYER). This proposal should be read in conjunction with the corresponding Commercial Proposal, which contain the pricing and commercial terms.

HIL is offering a solution that we are confident meets Buyer's and will bring significant value for Plant Managers.

It also sets the stage for a phased Digitalization, IT & business process maturity program, AI based projects, Centralised Plant Dashboard which aims at bringing alignment across the business & manufacturing processes.

These advancements are envisioned for subsequent phases and are not included within the scope of this current proposal.

3. PLANT TECHNICAL DATA

The existing production units & equipment to be considered in the solution are as follows:

3.1 4-HI SKIN PASS MILL

This is a 4-HI Single Stand Non-reversing Skill Pass Mill. A coil car loads the coils from the saddles onto the pay-off reel mandrel. OD & width measurement system is added with Auto loading facilities along with Coil weighing arrangement in Uncoiler and Recoiler. The coil will be opened and the strip front end will be fed through the pinch roll cum Leveller unit for flattening the front end of the strip and is then threaded over the threading table up to the mill roll bite. Then the strip is fed through the delivery side equipment upto the Re-coiler. The strip is then gripped in the Re-coiler, mill setting is done and then the tension is applied. On completion, the coil is usually taken out from the Recoiler.

Present automation level in the plant such as Level-1 system details, external systems/interfaces, and other plant/process information as required during design & development need to be provided by Buyer. Following is the plant technical information provided by Buyer as of now.

Main Specifications as provided by Customer:

S. No.	Item	Material Specification	Product
1	Material	Hot rolled, pickled Low carbon steel CQ & DQ	
2	Strip Gauge	Minimum: 1.0 mm Maximum: 6.0 mm	Elongation: Upto 4mm -15% Above 4mm – 10%
3	Strip Width	Minimum: 700 mm Maximum:1700 mm	Minimum: 700 mm Maximum:1680 mm
4	Coil Diameter	I.D.: 610mm762mm with Steel Rider O.D.: Max. 2,200mm Min. 900mm	I.D.: 610mm O.D.: Max. 2,200mm Min. 700mm
5	Unit Coil Weight	Maximum: 22kg/mm	Maximum: 22kg/mm
6	Coil Weight	Maximum: 36,000kg	Maximum: 36,000kg

Mill Specifications details:

a. Mill Type: 4-HI Single Stand Skin Pass Mill

b. Mill Roll Size (nominal diameter x barrel length)

   i. Work Roll: 480-525mm-R5 dia. × 1,800mm Length

   ii. Back -up Roll: 920-1250mm dia. × 1,750mm Length

c. Hand of mill: First pass (POR pass) from Left to right from operator side view

d. Rolling Speed: Max. 500m/min

e. Feeding Speed: Approx. 30m/min

f. Roll Separating Force: 1800 tons max

g. Strip tension of Tension Reel & Pay off Reel

   i. Pay off Reel: 18,000 Kg @ 250m/min
      14,500 Kg @ 500m/min

   ii. Recoiler: 26,500 Kg @ 250m/min
       20,400 Kg @ 500m/min

h. Motor Capacity

   i. Mill Motor: AC 1,500kW

   ii. Recoiler Motor: AC 1,690kW

   iii. Pay off Reel Motor: AC 1200kW

3.2 REWINDING LINE

S. No.	Item	Material Specification
1	Material	Cold Rolled Low Carbon Steel (Full Hard)
2	Strip Gauge	Minimum: 0.12 mm Maximum: 1.2 mm
3	Strip Width	Minimum: 900 mm Maximum:1680 mm
4	Coil Diameter	I.D.: 508mm/610mm O.D.: Max. 2,200mm Min. 900mm
5	Unit Coil Weight	Maximum: 22kg/mm
6	Coil Weight	Maximum: 36,000kg
7	Line Speed	500 m/min
8	Threading Speed	15 m/min

3.3 EXISTING AUTOMATION SYSTEM

S. No	Production Unit	Existing HMI & PLC Make & Model	Comm. Interface
1	SPM	**PLC**: ABB (ABB PM862)	VIP OR TCP/IP
2	Rewinding Line (3 Nos.)	**PLC**: ABB (ABB PM573)	VIP OR TCP/IP

Interface protocol with existing L1 system & PMEI system to be decided along with Buyer during detail engineering.

1 *Interface with existing Level-3 system will be via DB-LINK or TCP/IP (to be decided along with Buyer during detail engineering.)*

4. FUNCTIONAL SPECIFICATIONS

4.1 INFORMATION

This proposal outlines the features, options, facilities and screens which will be offered in the Process Monitoring & Efficiency Improvement(PMEI) System.

The offered PMEI system includes:

a) Level-2 process control system with adaptive control for rolling force calculation only
b) Unified Gateway Solution (for system interface)
c) Process efficiency monitoring
d) Delay management
e) Interface with L1/L3
f) Email alert and reporting system
g) SMS alerts
h) Remote support for six (6) months

4.2 GENERAL DESCRIPTION OF PMEI SYSTEM

The PMEI system is planned to be installed as a framework for centralized data storage in the on-premises data server for the CRM facility (4-HI SPM & Rewinding Line) at Jindal Steel Odisha Limited. Data collection to a centralized server and linking data from various sources meaningfully based on process logic/business logic is the first step towards digital transformation. Once the data is collected to a centralized database, it provides a platform for drawing data-driven inferences/insights. These insights help line managers, operation supervisors, top management in better and fact-based decision making. Effective visualizations, dashboards, alerts, and notification assist in quick and effective decision making.

1 The Unified Gateway solution from Seller will be used for collection and storing data from existing L1-System to the centralized on-premises PMEI server. The data in the PMEI system can be stored for maximum of up to 1 year. The number of tags that can be collected in PMEI server from the existing L1 system are 2800. Additional tags (over and above 2800) can be sent to PMEI server by addition of more tags in the unified gateway solution. The PMEI Application Software has the following functions:

a) Primary data input (PDI)
b) Mill set up function
c) Production Data Output (PDO)
d) Process data gathering
e) Roll administration
f) Data link with Level-3 computer
g) Data link with Level-1 PLC
h) Reporting Function
i) Downtime Management

4.2.1 Primary data input:

This is the basic function to input the technical information related to the coil for production. The PDI (Primary Data Input) information shall receive from LEVEL-3 system before coil arriving to the conveyor or skid of Cold Rolling Mill.

The main functions are:

a) PDI On-line receive from LEVEL-3

b) PDI manual input through HMI

c) Modify/Delete PDI data

d) Changing of production sequence

As all preset data for the mill control is based on this PDI information, received PDI information is checked against upper/lower values, data congruency, etc. and alarming to the operators if wrong data is found. The operator can correct the information in the PDI before rolling.

PDI display screen has provision for manual input of PDI and the rolling sequence in case BUYER does not have a Level-3 system.

Pulpit Operator can monitor PDI to cross check the actual coil in the field:

1) PDI of the coil currently in the Entry tension reel

2) PDI of the next coils on the conveyor or skid.

The contents of the PDI are basically the following. Detail definition and formats for the contents will be finalized at design stage.

a) Coil identification number

b) Entry gauge

c) Delivery gauge

d) Coil width

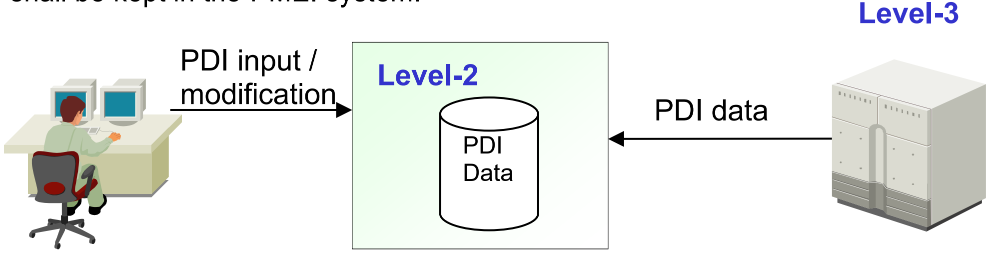
e) Material specification

f) Next processing (for example, annealing, etc.)

g) Coil weight

h) Other information

The PDI data structure is defined during design phase. PDI data for the coils on the conveyor and Mill shall be kept in the PMEI system.



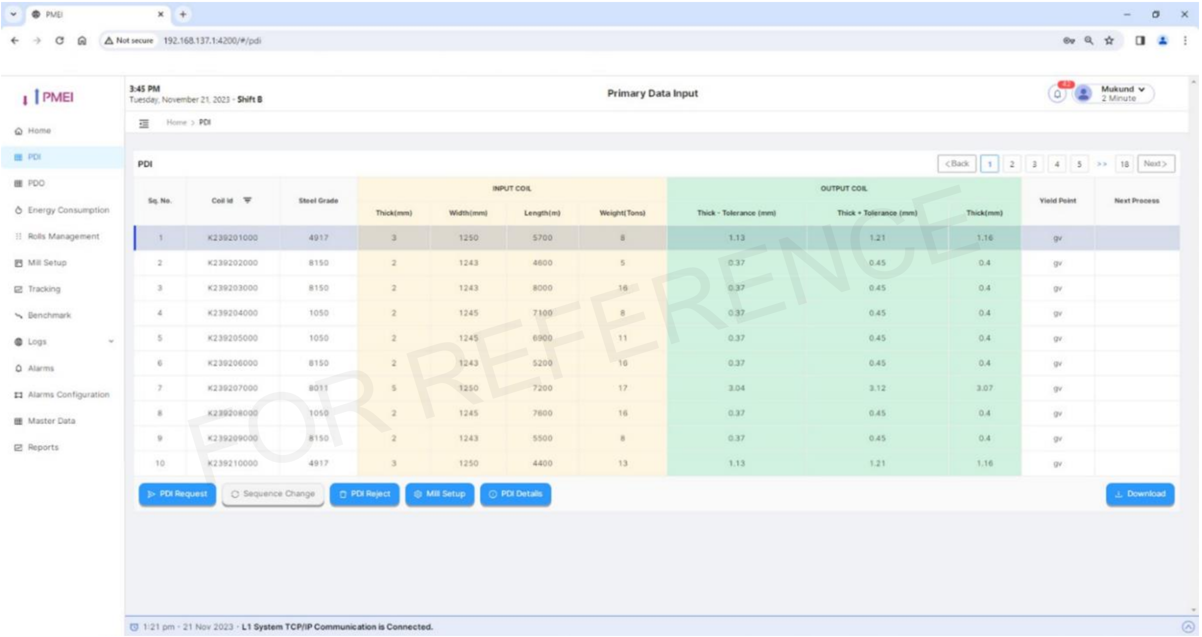


Fig-1: PDI data Transfer from Level-3

4.2.2 Mill Setup Function:

One of the functions of the PMEI system is to prepare Rolling Schedule for the mill according to the PDI and preset the reference control values for the PLC to achieve the desired product coil gauge.

The rolling schedule is initially decided by already existing recipe-based system and then Hitachi's adaptation system utilizes collected data from actual process to fine tune set-up values. Buyer shall provide access to existing recipe data functions, procedures, data tables, look-up tables and algorithms to enable this fine tuning.

The initial rolling schedule itself does not represent 100% of the physical rolling phenomena; therefore, it is complemented by such learning compensation called "Adaptive Control". The rolling schedule provides values as per look-up table provided by the Buyer.

The rolling schedule provides values of:

a) Mill speed

b) Tension (Back & Front Tension)

c) Pass reduction

d) Total reduction

e) Rolling Force

f) Mill Power

g) Others (details shall be finalized at design stage)

4.2.3 Adaptive control Function (Applicable only for SPM line):

Initial basic rolling schedule will be received from existing look-up table. PMEI system then uses this information to fine tune and optimize rolling schedule set-points using actual process data for optimization. This fine tuning and optimization of set-points is done using adaptive control.

The actual rolling results may differ from the basic rolling schedule calculated by from the existing look-up table. This might be caused by gradual changes in the mill dynamics and/or deviation of the coils from the nominal characteristics. Adaptive function is a supplementary function to the basic rolling schedule. This function is required to cover the deviations originated by:

a) Un-accounted variation of the parameters for basic rolling schedule.

b) Gradual changes of rolling dynamics of the plant.

The calculated parameters are compared with the actual data. The deviation values resulting from this comparison is used to calculate the adaptive compensation coefficients. This adaptive function is applied to rolling force calculation only. If BUYER find this feature valuable and are interested, it can be extended to other rolling parameters against a separate commercial contract.

The adaptive coefficients are stored in tables after statistical processing.

The adaptive control learns the mill behaviours from the deviation between initial setup values and the actual rolling results. The adaptive control updates the adaptive parameters (learning parameters) for the existing look-up table generated rolling schedule. The basic updating procedure is as follows.

Adaptive procedure

Data acquisition (Actual mill operation result)
Items: Entry thickness
Delivery thickness
Roll speed
Front tension
Back tension
Roll force

↓

Calculate the roll force using the measure data.

↓

Update the adaptive coefficient using the differences

↓

Store the adaptive coefficient in the statistical table

Fig-2: Adaptation Control Workflow (For reference only)

4.2.4 Delivery File Control

The main job of the Delivery file control function is to make product coil file at the coil unload event. At unload timing, PMEI system makes product coil file (PDO: Production Data Output). The PDO data is sent to the Level-3 system after each coil is unloaded.

4.2.5 ROLL MANAGEMENT

The Roll Data Management module within the PMEI system provides a comprehensive and reliable solution for monitoring, updating, and coordinating mill roll operations. It ensures seamless integration with upstream and downstream systems, enabling precise coordination across production stages.

Key Features:

a. Automated Roll Registration
The system efficiently captures and registers new roll data upon receipt from L3/RMS systems or manual operator input, enabling real-time updates and minimizing manual intervention.

b. Operational Roll Categorization: Manages current & next roll data based on usage conditions. For current rolls, the system tracks cumulative rolled length and weight, supporting further performance analysis of rolls.

c. Event-Driven Roll Transitions: Responds to roll change events initiated by operator requests through existing L1 system. Upon roll change event, next roll information is sent to L1 system & current roll data (rolled weight, rolled length etc.) is sent to L3/RMS system.

d. Production Data with Roll information: Production Output data (PDO) is sent to L3 system after each coil is produced. The PDO data also contains the roll information of the rolls used.

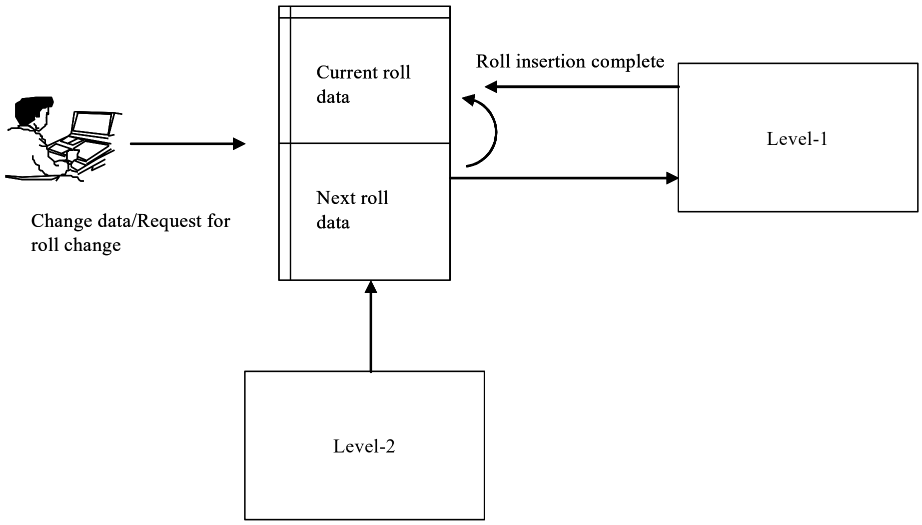


Fig-3. Mill roll management function concept

4.2.6 Email Alert and Reporting System

This feature will send production reports, log data, etc. to specified persons at a predefined frequency and timing. Detailed contents on this reporting feature will be finalized during engineering discussion and mutually agreed.

Automating reporting system can deliver substantial benefits to customer. Reporting system automation will reduce the time spent reporting and allow your team to do more process optimization and revenue generating activities. Reporting and email alert system ensures that you can deliver more comprehensive and customized reports to key executives, more quickly.

Email reporting feature is provided with PMEI System as a complimentary feature.

"Essential network configurations, tasks involved in connecting to the network, monitoring the network connection (including connection changes), giving users control over an app's network usage and authentications are to be decided and made/managed by buyer."

4.2.7 Remote System Support

SELLER will provide complimentary remote maintenance support for troubleshooting and issue resolution (if any) from the date of completion of project for first six (6) month period. Necessary infrastructure and network configuration will have to be enabled by BUYER to facilitate this. Remote maintenance support will be limited to scope of supply of this proposal. Remote support does not include additional features, modifications, and requests; its scope is limited to data gathering, system fine tuning, troubleshooting and issue resolution. Additional requests can be handled separately and not part of this support.

The effective working hours at HIL Office shall be from 9:00 AM to 5:00 PM IST (Monday to Friday), excluding National Holidays.

NON-DISCLOSURE AGREEMENT (NDA) will be signed between Buyer and Seller before starting the remote support.

4.2.8 Documentation Control

SELLER will provide documentation with respect to general functional specifications, hardware and software specifications, HMI functions, Wiring diagram, System Utility Document, interface documents.

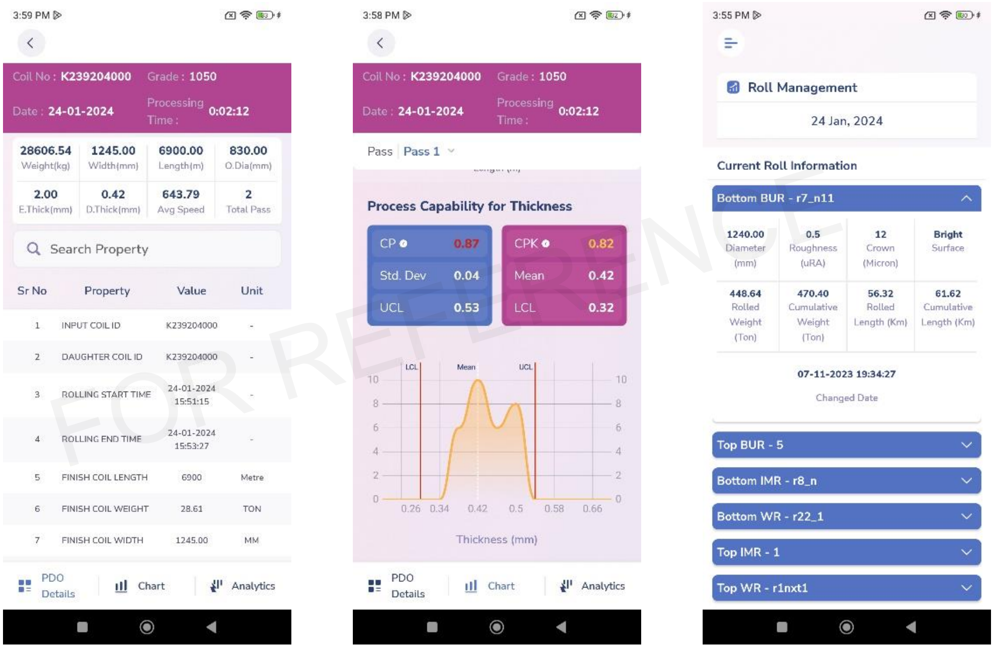
4.2.9 Customer Training

SELLER will provide hands-on training on usage of system from operation perspective during commissioning to maximum of 5 persons for maximum of 3 days. Training shall cover mutually agreed topics from the scope of supply.

5. VALUE-ADDED OFFERINGS

5.1 MOBILE APPLICATION

This is an android-based Process Monitoring Application, which allows senior executives of plant to have access to process data by their mobile phones by providing necessary access and authentication. Using this application, process parameters data, historical data for various coils can be viewed by specifying appropriate keys. Essential network configurations, tasks involved in connecting to the network, monitoring the network connection, giving users control over an app's network usage and authentications are to be decided and made/managed by buyer. Detailed contents on the application will be finalized during engineering discussion and mutually agreed.



<u>Fig-4: Mobile Application Sample Screen (For reference only)</u>

5.2 UNIFIED GATEWAY SOLUTION

PMEI system provides an integrated, end-to-end structure for data centralization in production process while providing a platform for unit's supervisors to execute manufacturing operations and planners to plan collaboratively. At the same time, it monitors and aggregates the performance indicators of rolling process.

<u>The proposed system uses a Unified Gateway solution</u> (as shown in the figure below) which is used for integrating core control and communications of automated machines and production facilities. The most efficient way Buyer can successfully manage the process and reach their strategic organizational goals is by measuring, collecting, analysing important KPIs, thereby realizing their digital benefits every step of the way.

The unified gateway solution provides seamless connection between control layer and information layer. Using these, it is possible to collect data from existing facilities using various open networks and communication protocol. Data collected is stored as digital data, making it possible to visualize and have on-site maintenance.

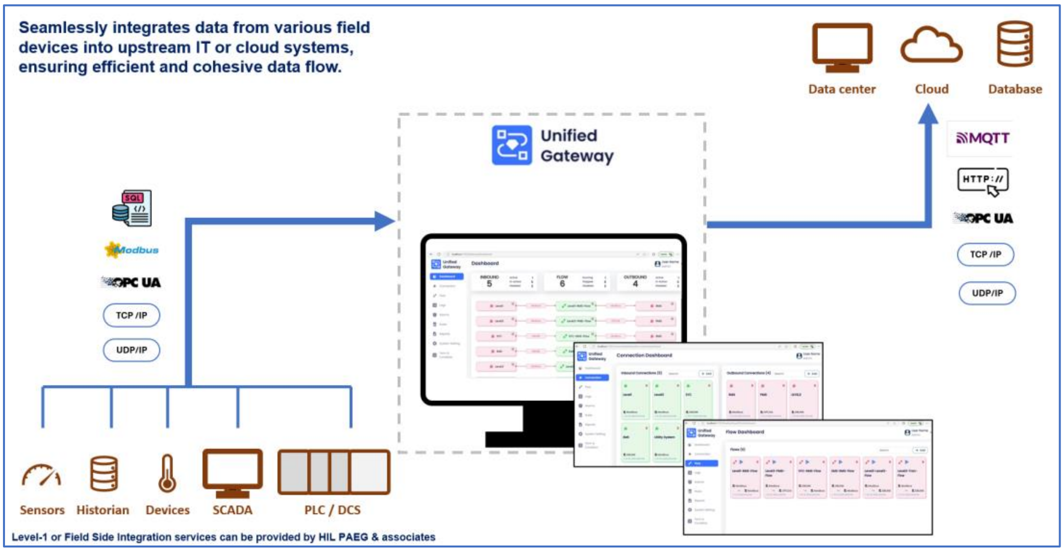


Fig-5: Unified Gateway Solution Description (For reference only)

5.3 PROCESS MONITORING & EFFICIENCY IMPROVEMENT

5.3.1 Centralized data collection

PMEI system will collect the data from existing databases of Mill L2 System, Mill L1 System & Rewinding L1 System. The proposed system can store coil production data for last 6-months. The maximum total number of tags that can be collected in the Unified Gateway System are limited to 2800.

5.3.2 Data Visualization Dashboard

Data Visualization dashboards and analytics dashboards will be provided and detailed contents/formats/structure on the dashboard will be finalized during engineering discussion and mutually agreed. Visualization dashboards can be fully customized as per buyer requirements after mutual discussions.

Data visualization is based on the simple proposition that it is easier to see connections, relationships, and trends when data are graphically displayed.

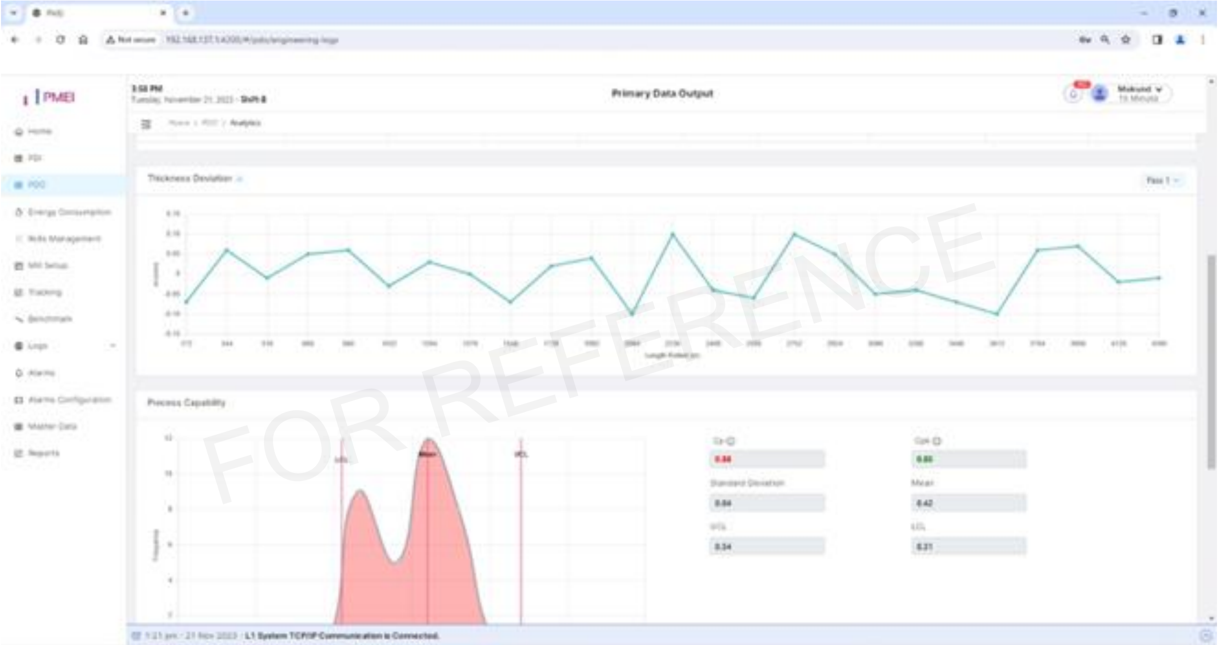


Fig-6: PMEI Sample Screen (For reference only)

5.4 SMS ALERT SYSTEM

It is important to take actions/decisions quickly to have desired effect and reduce unwanted productivity or quality loss. For this purpose, SMS alert system provides real-time updates to plant executive about significant events in the processing line. Contents of SMS and which events should trigger SMS alerts can be discussed and mutually agreed during detailed engineering discussion.

5.5 EMAIL ALERT AND REPORTING SYSTEM

This feature will send production reports, log data, etc. to specified persons at a predefined frequency and timing. Detail contents on this reporting feature will be finalized during engineering discussion and mutually agreed.

Automating reporting system can deliver substantial benefits to buyer. Reporting system automation will reduce the time spent reporting and allow your team to do more process optimization and revenue generating activities. Reporting and email alert system ensures that you can deliver more comprehensive and customized reports to key executives, more quickly.

The email reporting feature is provided with the PMEI system as a complementary feature.

“Essential network configurations, tasks involved in connecting to the network, monitoring the network connection, giving users control over a script network usage and authentications are to be decided and made/managed by buyer."

6. TECHNOLOGY STACK

The technology stack for various components is as follows:

Components	Technology Used
Application Front-end	Angular Framework. Application can be viewed on a standard web browser like Chrome, Edge & Mozilla
Database	Existing Database
Application Back-end	Python Django
Docker	For Deployment

6.1 BASIC HARDWARE SPECIFICATIONS

ELLER shall consider using latest version before detail design start. SELLER will submit documents to be approved by the buyer. IP address series for the network shall be decided mutually by Seller & Buyer. Also, domain should be clear while installing the software. Conditions for performance acceptance test are as per APPENDIX-I.

No.	Equipment name	Maker / Type	Quantity Main	Specification
1	PMEI Server	HP/Dell/Lenovo Enterprise Server (Tower based) Or equivalent	2	Intel Xeon-Gold Processor 32 GB DDR4 RAM 1 TB Memory 4 Ethernet LAN ports (1Gbps) USB Keyboard and mouse
2	PMEI Server Console	HP/Dell/Lenovo Monitor	2	FHD (1920 x 1080) resolution, 24 inch monitor
3	GSM Modem	Dynalog or Equivalent	1	
4	LIU	Any reputed make	1 Set	Standard
5	Media Converter	Digisol / D-Link / etc.	1 Set	FO to Ethernet
6	Ethernet Switch (managed)	HIL Approved	1	
7	Ethernet Switch (unmanaged)	HIL Approved	1	
8	Ethernet Cable (200m length)	Any reputed make	1 Set	
9	FO Cable (600m length)	Any reputed make	1 Set	
10	PMEI Client	HP/Dell/Lenovo Desktop PC	4	8 GB RAM, 500 GB HDD
11	PMEI Client Console	HP/Dell/Lenovo Monitor	4	FHD (1920 x 1080) resolution, 24 inch monitor

6.2 BASIC SOFTWARE SPECIFICATIONS

The PMEI system will be offered under “Python+Django” platform. SELLER shall consider using latest version before detail design start. The software specifications for “Python+Django” environment is as under:

No.	Name	Maker	Q'ty
1	Windows Server 2022 Enterprise 64bit	Microsoft	2
2	Python	-	2
3	Angular Framework	-	2
4	Microsoft Office Standard 2024	Microsoft	2
5	MS SQL Server 2022 Express Edition OR other suitable DB	Microsoft or SELLER recommended	2
6	Django Framework	-	2
7	Docker	-	2
8	Apache server version 2	-	2
9	Antivirus Software	TrendMicro	2
10	Windows 11 OS	Microsoft	4

6.3 THIRD PARTY SOFTWARE REQUIREMENTS

Database: Existing Database of the plant will be reused.

Remote Link: To provide a suitable level of response to operation & process execution problems and queries raised on site, Seller requires a network connection via broadband / VPN to be established between Seller and the Buyer’s site. Set-up and maintenance of this link will always be under the BUYER’s authority.

6.4 SYSTEM ARCHITECTURE

Overview of the Process Monitoring & Efficiency Improvement reference architecture is shown below:

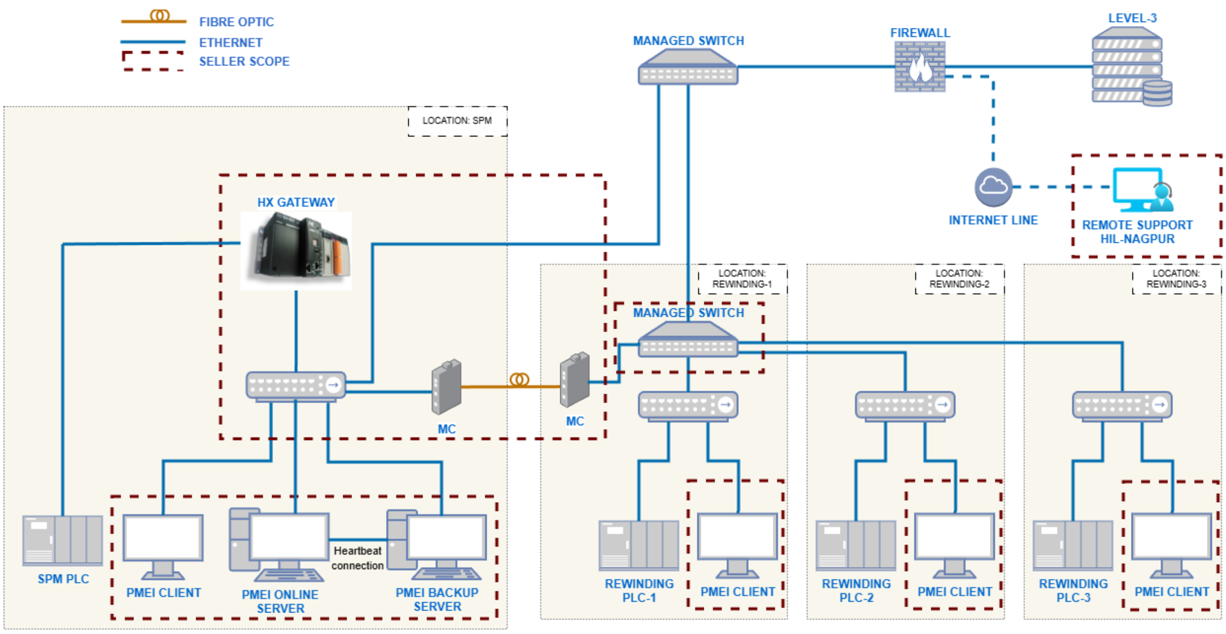
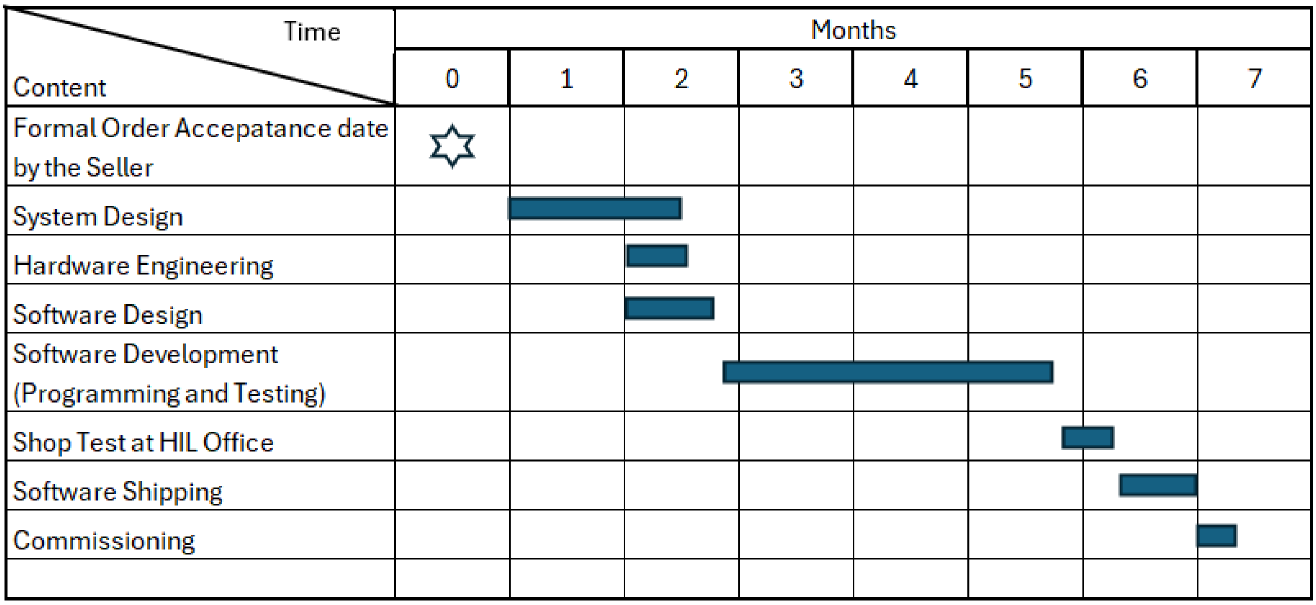
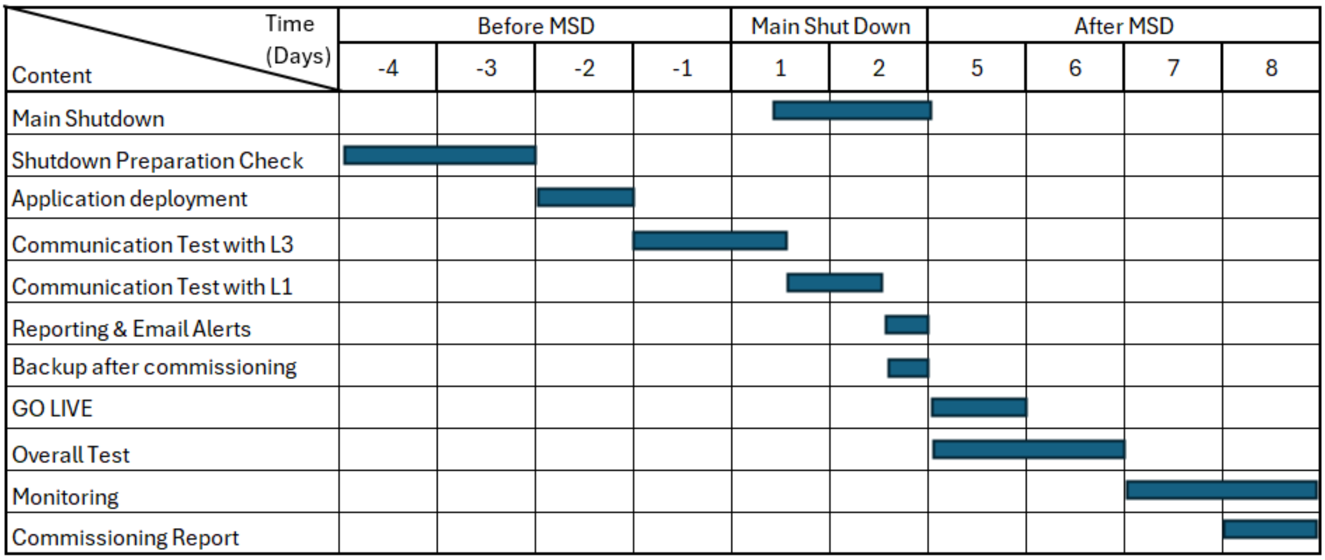


Fig-6: System Configuration (For reference only)

6.5 OVERALL SCHEDULE INDICATIVE FOR REFERENCE ONLY



6.6 SHUTDOWN SCHEDULE INDICATIVE FOR REFERENCE ONLY



The above shutdown schedule provided is for reference only. The final shutdown schedule will be determined through discussion and mutual agreement between the BUYER & SELLER.

The site-supervisor from seller's side will be deputed to the site for the commissioning, deployment & training at site:

Total 20 working days (Inclusive of on-site training).

Travelling days are not included in the above days.

7. SCOPE OF SUPPLY

7.1 SCOPE OF SUPPLY DEFINITIONS

The engineering and software development service necessary for the proposed PMEI system shall be provided either by Buyer or by the Seller in accordance with the Scope of supply List. The matters not described in the said list shall be determined through mutual discussion.

Buyer : Jindal Steel Odisha Limited (B: Buyer)

Seller : Hitachi India Pvt Ltd (S: Seller)

BD : Basic Data
BE : Basic Engineering
DD : Detail Design
SU : Supply or Service
ER : Erection
COM : Commissioning

X/Y: Primary Work/Services provided by X with Support from Y

HITACHI

7.2 DIVISION OF SUPPLY, ENGINEERING AND SERVICES

No.	ITEM	Responsibility					
		BD	BE	DD	SU	ER	COM
(1)	**Services**						
-1	Project Execution	B/S	B/S	B/S	-	-	-
-2	Overall system design	S	S/B	S/B	-	-	-
-3	Work Test and Simulation	S	S	S	-	-	-
(2)	**System Engineering**						
-1	PMEI Hardware design	S	S	S	-	-	-
-2	PMEI Network Design	B/S	B/S	B/S	-	-	-
(3)	**Hardware**						
-1	Managed Switch (1No.)	S	B	B	S	B	B/S
-2	Unmanaged Switch (1No.)	S	B	B	S	B	B/S
-3	Rack for Network Switches / HUB.	B	B	B	S	B	B/S
-4	Ethernet cables	B	B	B	S	B	B/S
-5	PMEI Server	S	S	S	S	B	S
-6	HX Gateway	S	S	S	S	B	S
-7	Desk for Server	B	B	B	B	B	B
-8	Client PCs (Maximum 4Nos.)	S	B	B	S	B	B/S
-9	GSM Modem	S	S	S	S	B	B/S
-10	SIM for GSM Modem	B	B	B	B	B	B
-11	Firewall	B	B	B	B	B	B
(4)	**Software**						
-1	Section 6.2	S	S	S	S	-	S
(5)	**Interface with External Systems**						
-1	Interface with L1 system (at L1 side)	B/S	B	B	B	-	B/S
-2	Interface with L1 system (at L2 side)	B/S	S	S	S	-	B/S
-3	Interface with L3 system (at L3 side)	B/S	B	B	B	-	B/S
-4	Interface with L3 system (at L2 side)	B/S	S	S	S	-	B/S
-5	Email Server SMTP (for email reports)	B	B/S	B/S	B/S	-	B/S
(6)	**TRAINING**						
-1	Training (Max 3 days) for 5 persons	S	S	S	S	-	-
(7)	**CONSTRUCTION & ENGINEERING**						

-1	**Cables (Wherever Required)**						
	Power cables	S	B	B	B	B	-
	LIU & Media Converter	B/S	S	S	S	B	B/S
	Patch Cords (Ethernet & FO)	S	S	S	S	B	B/S
-2	**Erection Materials**						
	Power and control junction boxes	B/S	B	B	B	B	-
	Cable Trays, conducts, lugs, Glands & other cabling material	B/S	B	B	B	B	-
-3	**Earthing System**	B/S	B/S	B	B	B	-
(8)	**Pre-Engineering Activities**						
-1	Study of Existing system	B/S	B/S	B/S	-	-	-
-2	Design of L1 - PMEI interface	B/S	B/S	B/S	-	-	-
-3	Design of L3 - PMEI interface	B/S	B/S	B/S	-	-	-
(9)	**Documents to be submitted for Reference & Records**						
-1	Functional Specification	S	S	S	S	-	-
-2	Screen Design Document	S	S	S	S	-	-
-3	Hardware Specification	S	S	S	S	-	-
-4	System Interface Document	S	S	S	S	-	-

Note: Any additional requirements beyond the scope mentioned above shall be discussed and mutually agreed upon. A separate proposal will be submitted for such additional requirements.

7.3 WORK COMPLETION CERTIFICATE

Upon the successful completion of the work (based on the below criteria), the Seller shall issue the Work Completion Certificate to the Buyer.

Work Completion Criteria:

- Supply of Hardware & Software as per the scope of supply (described in section 6.1 & 6.2)
- Completion of data collection from the existing L1 System to the centralized PMEI server.
- Completion of Commissioning works Man-days (as described in section 6.6)
- Submission of all documentation as per the scope (described in section 7.2 (9))

Upon completion of the above criteria, the Work Completion Certificate will be issued by the Buyer to the Seller, signifying the satisfactory completion of the project. The Buyer shall issue the Work Completion Certificate within seven (7) days of the completion of the above criteria.
In case if the Work Completion Certificate is not issued by BUYER within this period, it is considered that Work Completion Certificate as deemed issued by the BUYER. If there are any issues/problems faced by BUYER, it will be mutually discussed, and SELLER would support BUYER with the best of the ability to resolve the issue. By issuing the Work Completion Certificate, the Buyer acknowledges that the Seller has fulfilled all contractual obligations and requirements.

7.4 BUYER OBLIGATIONS

The Buyer should fulfil the following obligations

- Responsible for the project execution (answer technical queries in reasonable time and coordinate with all the stake holders of the project)
- Arrange all the hardware in Buyer scope well ahead of the agreed time schedule.
- Network cables & accessories not included in this proposal to be provided by Buyer
- Site access to Seller's representative for data collection and discussion with the technical team as per requirement.
- Dedicated internet connection
- In case of any health issue to Seller's representative, buyer to immediately provide best available medical facility. The expenses will be borne by the Seller.
- Latest Level-1 Software backup along with last 3-month Coil production data & DB Schema for Level-3 to be provided to Seller
- If required, availability of 3rd party equipment engineers at site for engineering discussions and commissioning.
- Buyer will provide interface to the existing look-up tables and/or mathematical models for information exchange with PMEI system. The detailing of this shall be done during detail engineering stage.
- Essential network configurations, tasks involved in connecting to the network or internet, monitoring and configuration of the network for email alert & SMS alert will be done by buyer.

7.5 EXCLUSION LIST

Only the services mentioned in the Scope of this proposal are included. Any further services need to be mutually discussed, and commercial / technical implications discussed and compensated through a previously agreed change management mechanism.

Specific attention is drawn to the following services that are excluded from the scope of this proposal (unless specifically included elsewhere in this proposal).

In this proposal following are not included:

- Interface with external devices other than explicitly described in the proposal document.
- Software patches including but not limited to Microsoft Security updates, Antivirus updates or any other software upgrades or patches.
- Any warranty, guarantee, liability, responsibility, etc. about productivity, quality, yield, etc.
- Any equipment, components, tools & works not described in this proposal.
- Source code of software of core technology.
- Modifications to existing electrical and automation systems at Mill including but not limited to existing Level-1, Level-2, MES, SYC, CYC system.
- Erection activities wherever required for project execution. Any kind of Civil work.
- Hardware and software other than that is mentioned in technical document.
- Mechanical equipment supply, modification etc.
- Support for troubleshooting for mechanical/operation/process issues of customers.
- PDA software and related hardware
- Firewall and other networking components
- Performance with respect to productivity, yield, quality, process capability, process performance, etc.

7.6 BINDING CONDITIONS

- SELLER scope of supply & services as described in agreed & signed technical specifications (TS) shall be limited, as per the inputs/information obtained from the BUYER, only to the modification/creation of interface program for helping the BUYER to use SELLER supplied equipment/items for obtaining the required functioning at BUYER own responsibility.
- All the services, except for those under the scope of SELLER as per the agreed & signed TS, will be under the full responsibility of BUYER. SELLER neither guarantees any performance nor SELLER is liable for any claims for the non-performance of such Equipment & services.
- SELLER scope of supply & services excludes any supervision, management, regulation, arbitration and/or measurement of BUYER's personnel, agents or contractors and work related thereto nor does include any responsibility for the scheduling, monitoring or management of their work.
- SELLER supervisors shall have the right to leave the site without any permission or approval of BUYER in the following cases:

  a. If the idling periods extend more than a considerable time as judged by SELLER.
  b. If the BUYER does not follow the conditions of the agreed terms as per the agreed and signed TS
  c. An act of God such as natural calamities, war, riot, rebellion, invasion, commotion, epidemics, terrorism, or such other phenomenon which is predictable to affect the body or life of our supervisors directly or indirectly.
  d. If BUYER prerequisites are not fulfilled as per the agreed and signed TS.
  e. Circumstances at project site are not conducive and are beyond SELLER’s control as judged by SELLER.

- SELLER will not be responsible for

  a. Performance guarantee of any kind.
  b. Productivity AND/OR Quality of the Product

- SELLER will not be responsible for any failed/missing/damaged equipment, item, and components etc. during the execution of the service work unless otherwise explicitly mentioned in the contract.
- Unless explicitly mentioned in the agreed and signed TS, SELLER will not take responsibility for any modification or updating of the existing system.
- The effective date of execution of the contract shall be the PO acceptance date by the SELLER.

7.7 CYBERSECURITY DISCLAIMER

SELLER strongly recommends that security updates are implemented by Buyer as soon as they are available and that the latest versions are used. Use of versions that are no longer supported, and failure to apply the latest updates may increase the exposure to cyber threats. This application contains confidential information. It is the sole responsibility of end-user to maintain the confidentiality and proper usage of the information contained herein. The BUYER is responsible for preventing unauthorized access to the existing plant systems. The BUYER will be responsible for Cybersecurity, Information Security, Hardware Security, Personal Data Security, Overall Data security, Network security, equipment security. In no event shall SELLER be liable for any direct, indirect, incidental, consequential, or special damages arising out of or in connection with any cybersecurity incidents, data breaches, unauthorized access, or any other security-related events, regardless of the cause of action.

8. DISCLAIMER

8.1 SOFTWARE LICENSES

The use of the delivered software, or parts of the same, is strictly limited to the services provided. The buyer has the right to use and modify the software for normal use and for the lifetime of the system supplied by Seller. Any duplication (except for normal back-up purpose), distribution to third parties, sell or use in other plants is prohibited. It's also prohibited to make software accessible to third parties. All rights of the software remain with Seller.
'BUYER is responsible for maintaining and ensuring valid license for use of third-party software and legal use of third-party software used in this system. In no event shall Seller be liable for any issue or claim or damage arising out of use of any third-party software.'

6.1 CHANGES DUE TO TECHNICAL IMPROVEMENTS

Seller follows a policy of continuous improvement of design, manufacturing and software development and therefore reserve the right to supply systems differing in detail from what is described in this document.

6.2 CONFIDENTIALITY OF INFORMATION (Non-Disclosure Agreement)

According to the law, Seller considered this document to be a company secret and therefore prohibits any person to reproduce or disclose it in whole or part to third parties, without specific written authorization of Seller Management.

6.3 LIMITATION OF LIABILITY & CONSEQUENTIAL DAMAGES

With respect to any and all liability of HIL out of or in connection with this proposal notwithstanding anything contained herein to the contrary in no event shall HIL's total liability for any and all claims, damages or liabilities arising out of related to its performance of this proposal, including but not limited to any liquidated damages, costs for repairs, replacement, rectifications of defects, dismantling, etc., exceed one hundred percent (100%) of the HIL's Contract Price. In no event shall HIL be liable for any indirect or consequential damages, including but not limited to any loss of profit, loss of use, loss of Contracts, loss of production or business interruption. According to law, SELLER considers this document to be a company secret and therefore prohibits any person from reproducing or disclosing it in whole or part to third parties, without specific written authorization of SELLER Management.

9. FREE PROOF-OF-CONCEPTS

Seller is pleased to provide a **complimentary Proof of Concepts (PoCs)** along with the aforementioned system to showcase SELLER's latest offerings in Composite Automation & Digital Solutions(CADS) & Digital Transformation (DX) Solutions. The PoC will be conducted on the same server.

The solutions under the PoC will be deployed for a period of 2 months to demonstrate the capabilities of these solutions. Upon successful completion of the demonstration, the BUYER must capitalize on the benefits by purchasing these solutions through a separate commercial contract, ensuring continued access to their full potential. The SELLER reserves the right to withdraw the PoC solutions if the BUYER does not proceed with the purchase following the demonstration.

9.1 SCHEDULE PLANNING SOLUTION

The schedule planning solution offers an advanced material and order allocation features. Users can allocate materials to orders based on real-time data and predefined rules. This ensures optimal utilization of resources, reduces waste, and enhances production efficiency. Material and order allocation will help user to achieve better coordination and control over the production process.

The system allows users to define basic scheduling rules to automate and optimize the production process by creating the production schedules automatically.

The system will enable the creation of primary data inputs (PDI) based on the specification data provided by Buyer. Users can input essential data required for production processes, ensuring accurate and reliable data management.

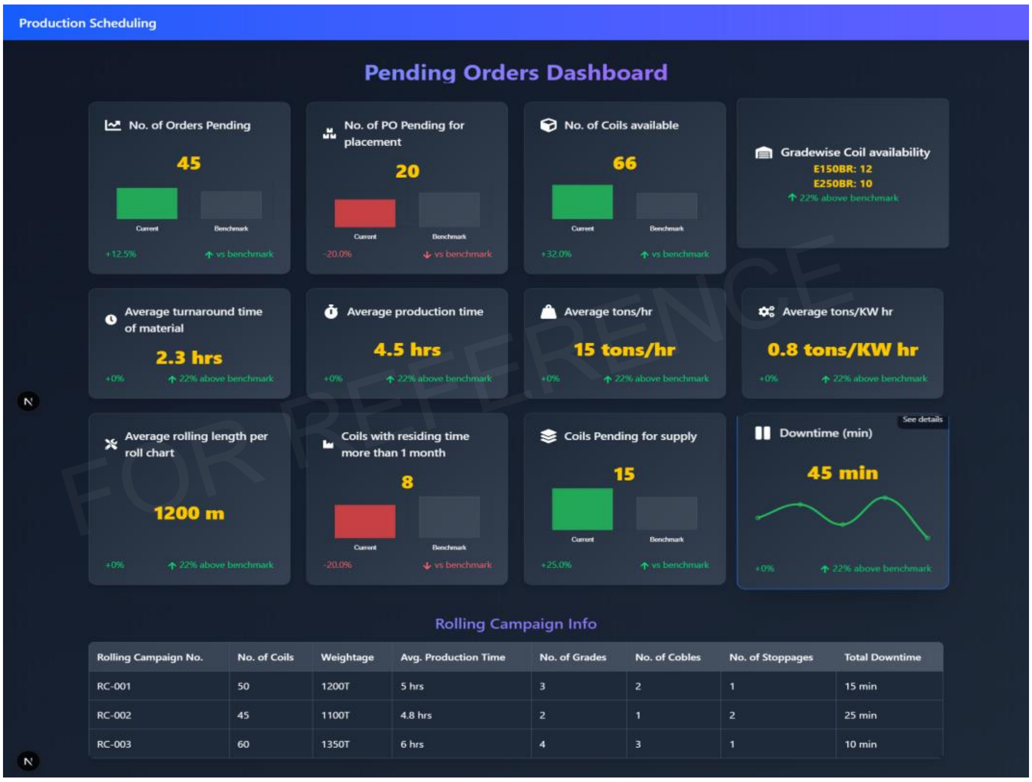


Fig: Production planning & Scheduling System Reference Screen

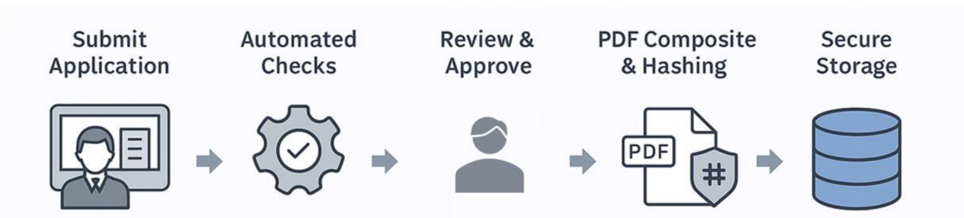
9.2 DIGITAL PERMIT SYSTEM

The current work permit process in industrial environments is largely paper-based, leading to delays, missing approvals, unauthorized modifications, and lack of centralized monitoring. To overcome these limitations, Seller proposes the Digital Permit System (DPS) – a robust digital solution designed to digitize, streamline, and secure the entire work permit workflow.

The digital permit system is a comprehensive digital platform that manages the entire permit application process from start to finish. Applicants can fill in their details and documents through a web interface, where the system performs automated checks before registering the application. Authorized approver from Buyer then reviews the submissions, request additional information if needed, and approve or reject the permits. Upon approval, the permit is digitally signed and compiled into a PDF. To ensure integrity, a **hashing technique** is used and stored along with the signed permit in secure databases within local infrastructure. The system also sends the signed permit via email to the applicant, with the issuing administrator copied for transparency. Email delivery is logged to maintain an audit trail and confirm successful transmission. The platform ensures document security through layered mechanisms including digital signatures, hashing, secure storage, and cross-verification via email.

Features:

- Mandatory fields prevent incomplete forms.
- No permit can move forward without required approvals/signatures.
- Unauthorized modifications are impossible once a permit is signed (hash + lock).



9.2.1 AUDIT TRAIL & TRACKING

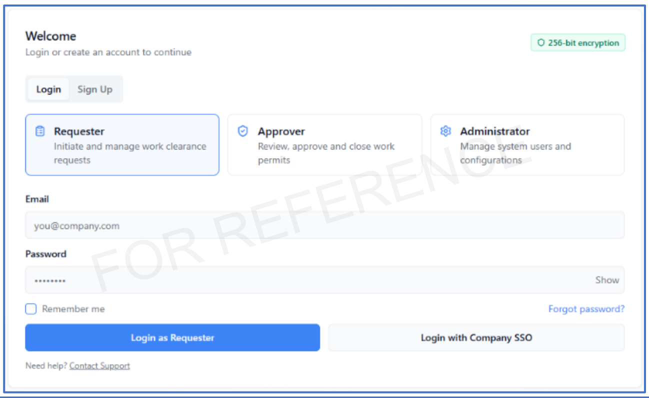
The system ensures complete transparency and accountability by maintaining a comprehensive audit trail of all user actions. Every activity—such as permit creation, modification, review, approval, rejection, or closure—is automatically time-stamped and tagged with the user's identity. In addition, the solution provides real-time status tracking, where each permit moves seamlessly through defined stages (Requested → Under Review → Approved/Rejected → Closed).

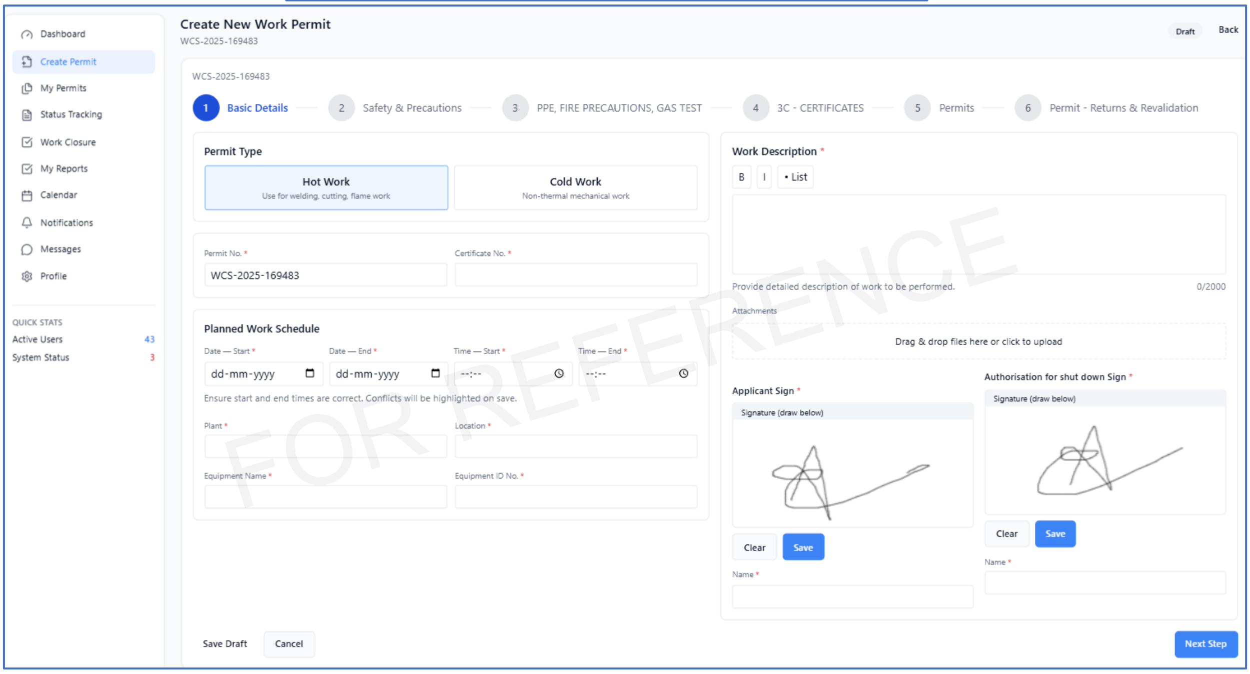
9.2.2 ENCRYPTION / HASHING & SECURE STORAGE

The system employs standard encryption and hashing techniques to ensure that every permit document remains tamper-proof. To strengthen data security, the signed permits and their hashes are stored in distinct, protected databases. This segregation adds a strong layer of resilience, ensuring redundancy and safeguarding against corruption or unauthorized access

9.2.3 CLOSURE OF WORK PERMIT

Once the assigned work is completed, the requester can formally submit a work completion request through the system. This request is then routed to the approver for review, who verifies and authorizes the closure of the permit. Upon approval, the permit is digitally closed and archived, ensuring that no permit remains open beyond its intended scope. This not only streamlines the end-to-end process but also supports effective audits and operational safety.





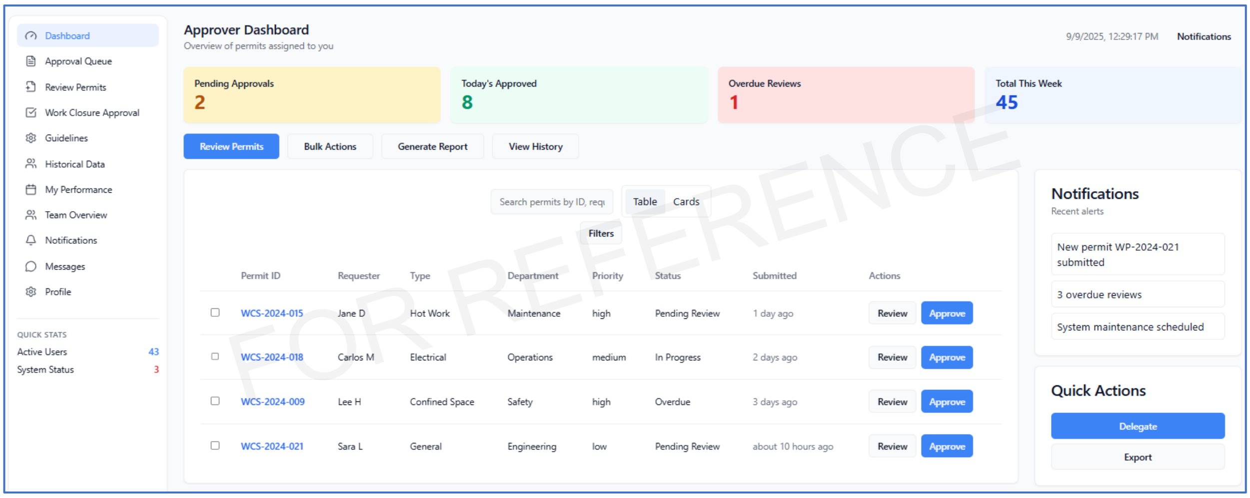


Fig: Digital Permit System Reference Screen

9.3 AUTOML PLATFORM

To drive Digital Transformation (DX) in the existing setup, Seller is pleased to propose the deployment of an “AutoML Platform” on the existing PMEI server for a limited period as a Proof of Concept (PoC).

The objective of this engagement is to demonstrate the platform’s ability to accelerate data-driven insights, streamline model development, and enhance decision-making capabilities without extensive manual intervention.

- The AutoML platform will be installed and configured on the existing PMEI server infrastructure.
- The PoC will run for a limited duration solely for demonstration purposes.
- The platform will enable:
  - Automated data preprocessing, feature engineering, and model selection.
  - Model evaluation, and performance benchmarking.
  - Hyperparameter Tuning
  - Demonstration of ML Model working through predictive and prescriptive insights.

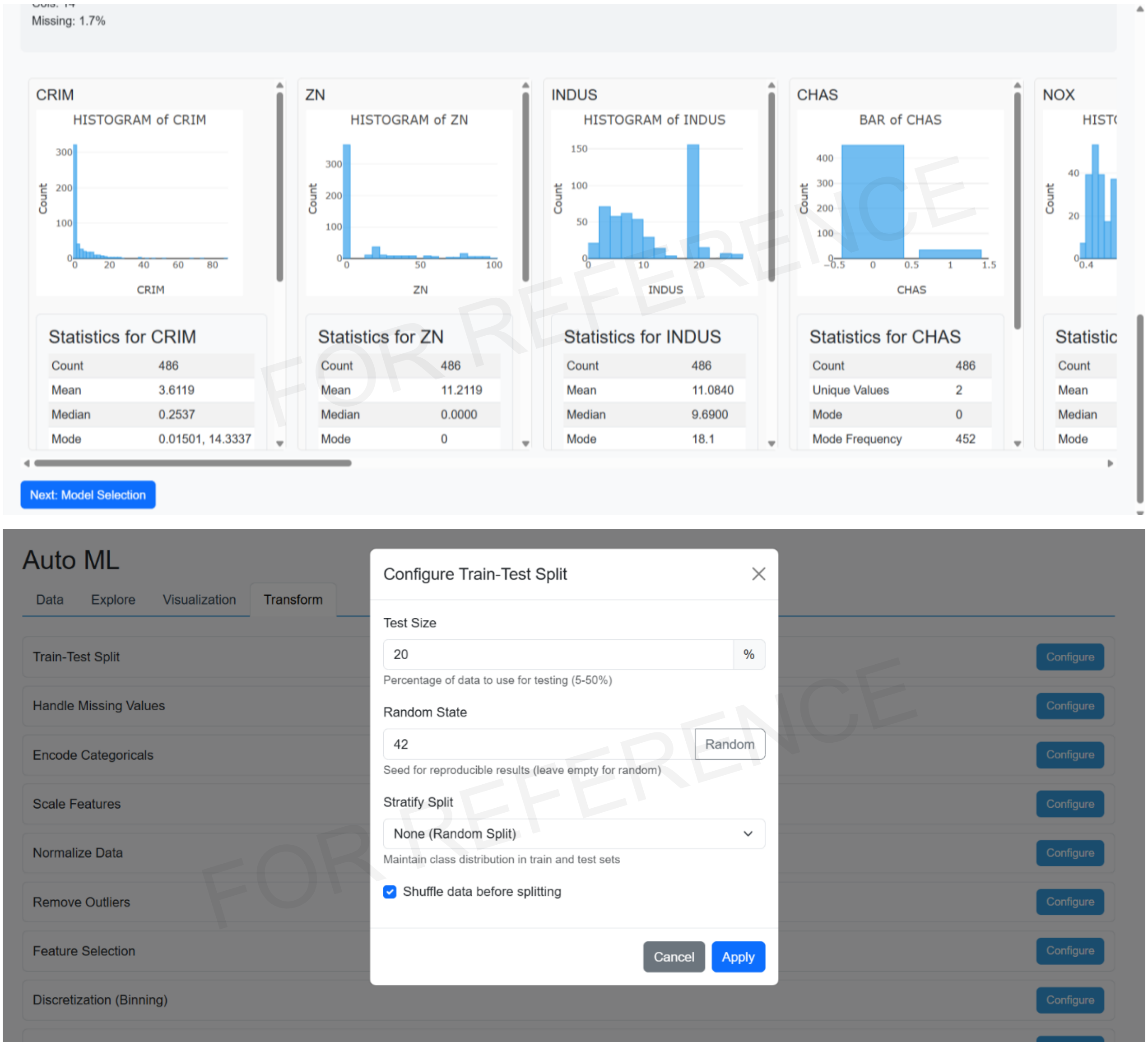


Fig: AutoML Platform Reference Screenshot

End of Technical Proposal